Visual contrast test results

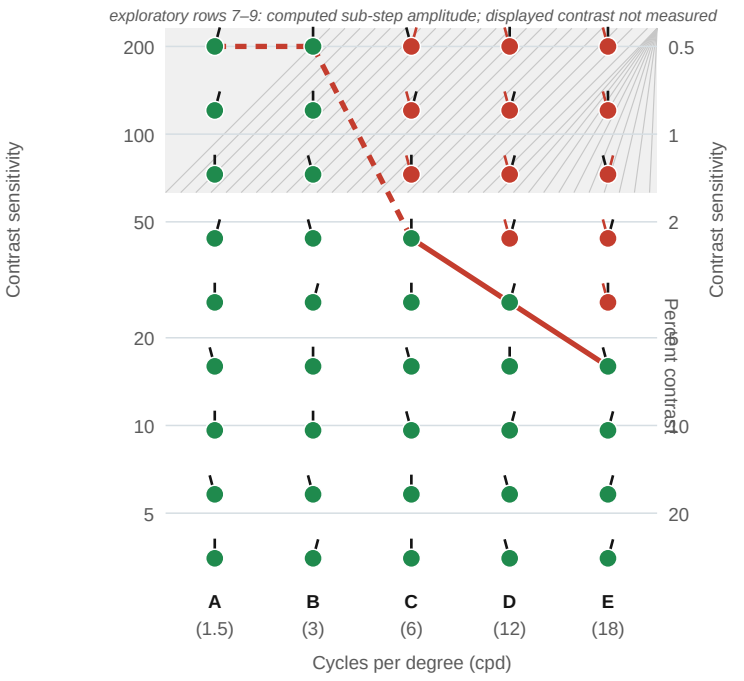
STAGING BUILD

Run: 3b1ba39e-ba6c-4d26-be03-56af0506e6b1
Protocol: full-5x9-v1 (every patch presented)
Geometry: 50 cm (typed) · 44.5 px/cm (card) · 38.84 px/deg · 2.16 px/cycle at 18 cpd
Completed: 2026-10-09T11:02:27.126+11:00
Name: _____
Status: COMPLETE_UNSAVED

Right eye 33 of 45 patches identified · Left eye 33 of 45 · Clinical significance not established

This record was not saved in the browser; keep the exported JSON file.

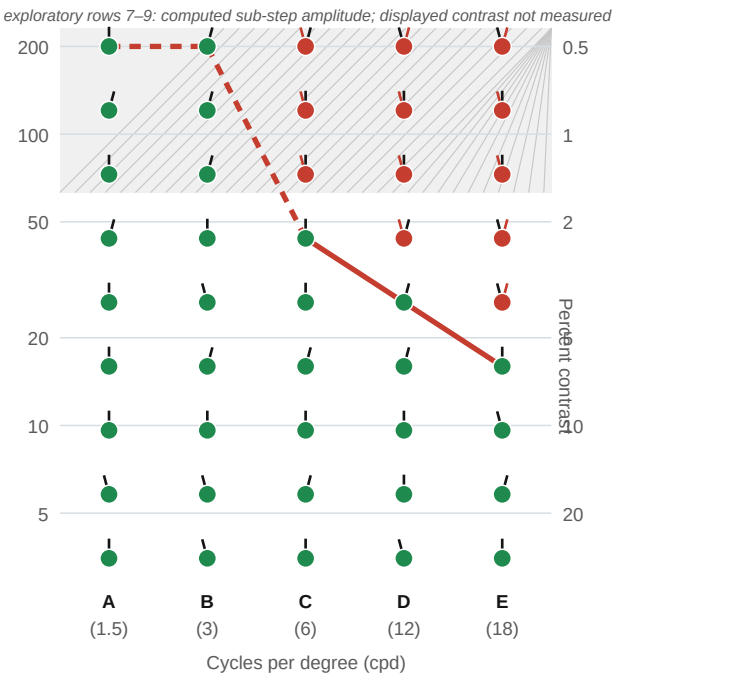
Left eye



33 of 45 presented patches identified

0 can't see · 0 not presented · 0 isolated

Right eye



33 of 45 presented patches identified

0 can't see · 0 not presented · 0 isolated

1. What was measured

This test showed you 45 circular patches of faint parallel bars for the right eye and 45 for the left, at five bar widths (1.5, 3, 6, 12 and 18 cycles per degree) and nine contrast levels from 28.6 % down to 0.5 %. For each patch you chose whether the bars tilted left, stood upright or tilted right. The charts and tables record exactly what you were shown and what you answered. They describe how faint a pattern you identified; they do not identify a cause.

2. Test conditions and their reliability

Screen size was set from a bank card (44.5 px/cm) and viewing distance entered as 50 cm, giving 38.8 pixels per degree. Brightness, colour settings and room lighting were not measured. The finest bars (18 cycles per degree) were drawn at 2.16 pixels per cycle. A sampled grating needs more than two pixels per cycle to be represented at all, and several more before its rendered contrast and shape approach the nominal values; at 2.16 pixels per cycle the column E patches were coarsely sampled and their effective contrast is uncertain. The three faintest contrast levels have, under the assumed rendering model (sRGB transfer function, 8-bit display, mid-grey background), computed amplitudes below one display step; what contrast those patches actually produced on this screen was not measured, and they are marked exploratory. Protocol: full (every patch presented). No change in browser zoom was detected during the test.

3. Results per eye and bar width

Each circle is one patch, placed by its contrast and bar width. Green: tilt identified. Red: not identified; the red tail shows the tilt chosen, the black tail the true tilt. Hollow: not presented (quick protocol). A ring marks an isolated correct response, one with a miss at an easier level. The hatched band covers rows 7–9, which are exploratory. The red line joins the lowest contrast correctly identified in each column and is dashed where it enters the exploratory band.

Right eye

	A (1.5)	B (3)	C (6)	D (12)	E (18)	Total
Presented	9	9	9	9	9	45
Identified	9	9	6	5	4	33
Lowest contrast identified	0.50 %	0.50 %	2.3 %	3.8 %	6.3 %	
Same, rows 1–6 only	2.3 %	2.3 %	2.3 %	3.8 %	6.3 %	

Left eye

	A (1.5)	B (3)	C (6)	D (12)	E (18)	Total
Presented	9	9	9	9	9	45
Identified	9	9	6	5	4	33
Lowest contrast identified	0.50 %	0.50 %	2.3 %	3.8 %	6.3 %	
Same, rows 1–6 only	2.3 %	2.3 %	2.3 %	3.8 %	6.3 %	

* isolated: the defining correct response has a miss at an easier contrast level.

4. Differences between the eyes

At each bar width the lowest contrast identified by the right eye and the left eye differed by 0.00 / 0.00 / 0.00 / 0.00 / 0.00 log units (a value above zero means the right eye identified a fainter patch). Differences of this size between a person's eyes can arise from the test's own variability; whether they are meaningful has not been established for this test.

5. Limitations

This is a self-administered screen on an uncalibrated consumer display. Results depend on the screen, its settings, the room, attention and guessing (one in three answers would be right by chance). Each contrast level was shown once per eye, so a single lucky or unlucky answer moves the result. The test has not been validated against clinical contrast sensitivity instruments, and no normal range has been established for it. Reduced contrast sensitivity, where present, has many possible causes, including uncorrected refractive error, cataract, glaucoma, macular disease, diabetes, medication, alcohol and fatigue.

6. When to consult someone

This test does not tell you whether your vision needs attention; an eye examination does. See an optometrist or your doctor if you have persistent difficulty seeing in dim light, fog, rain or glare, difficulty reading or recognising faces, any visual symptoms such as blur, haloes, distortion or a shadow in your vision, or any sudden change in your sight, which needs same-day attention. A consistent change across repeat tests taken under the same conditions is also worth discussing. Take this report with you; the numbers and charts are meant to be read alongside a clinical examination, not instead of one.

7. What this test does not show

A contrast sensitivity result on its own does not identify mould, mycotoxin or any other environmental exposure, and cannot diagnose any illness. Research has reported group-level associations between reduced contrast sensitivity and exposure to water-damaged buildings; it has also reported that the handheld chart version does not separate exposed people from controls, and expert bodies have concluded that no validated test for such exposure exists. Questions about your home or workplace are answered by an environmental assessment, and questions about your health by a medical practitioner.

Identification

Run 3b1ba39e-ba6c-4d26-be03-56af0506e6b1 · 2026-10-09T11:02:27.126+11:00 · Software v0.7.0 (staging) · Protocol full-5x9-v1 · Scoring score-v1-maxcorrect · Stimulus gauss-0.42R-sRGB-dither-v1 · Integrity SHA-256 30beeedfb0bb over test definition and responses (local integrity check, not proof of origin) · Geometry 44.5 px/cm (card), 50 cm (typed), 38.8 px/°, 2.16 px/cycle at 18 cpd, DPR 1 · Source c75633bdf485 · Commit f09b1506

Software verification: passed, suite phase0-T01-T28+phase1-M01-M13+guidance-G01-G06, v0.8.0, 2026-10-11 (53/53 tests), bound to source c75633bdf485. Physical measurement validation: not performed. Clinical validation: not performed; no diagnostic or exposure claim.

Evidence status

Visual contrast sensitivity measures how faint a pattern can be discriminated across spatial frequencies. It depends on both optical factors (refractive error, clarity of the cornea and lens, pupil size, light scatter) and neural factors (retina, optic nerve and visual cortex), so reduced sensitivity can arise from ocular disease such as cataract, glaucoma or macular disease, from uncorrected refractive error, and from neurological conditions, medication, alcohol and fatigue. Shoemaker popularised it as a bedside marker of biotoxin-related illness after exposure to water-damaged buildings. The evidence is mixed and the test is not validated as a diagnostic tool for mould-related illness.

A NIOSH study (ref 12) found lower VCS values at every spatial frequency among employees of a severely water-damaged school than at a comparison school, with more rash, respiratory and constitutional symptoms, yet advised against recommending VCS for mould-exposed people because earlier studies used non-representative samples, lacked comparison groups, characterised exposure poorly and were confounded by conditions such as fibromyalgia and chronic fatigue. A 2026 four-method comparison (ref 13) found the handheld Shoemaker chart did not separate exposed people from controls while digital tests did; the digital tools remain investigational and their figures do not apply to this software.

The American College of Medical Toxicology (ref 14) states there is no validated biomarker for mould or mycotoxin exposure; a Lancet Infectious Diseases review (ref 15) notes no diagnostic test exists for building-related illness; a proposed framework for dampness and mould hypersensitivity syndrome (ref 16) uses clinical and exposure criteria only. In an Australian clinical context VCS should be treated as non-validated and not a stand-alone diagnostic or monitoring tool.

References

1. Ginsburg AP. A new contrast sensitivity vision test chart. Am J Optom Physiol Opt. 1984;61(6):403-407.

2. Ginsburg AP. Contrast sensitivity, drivers' visibility, and vision standards. Transportation Research Record. 1987;1149:32-39.

3. Ginsburg AP. Contrast sensitivity and functional vision. Int Ophthalmol Clin. 2003;43(2):5-15.

4. Hudnell HK, House D, Schmid J, Koltai D, Stopford W, Wilkins J, Savitz DA, Swinker M, Music S. Human visual function in the North Carolina clinical study on possible estuary-associated syndrome. J Toxicol Environ Health A. 2001;62(8):575-594. doi:10.1080/15287390151079633

5. Shoemaker RC, Hudnell HK. Possible estuary-associated syndrome: symptoms, vision, and treatment. Environ Health Perspect. 2001;109(5):539-545.

6. Hudnell HK. Chronic biotoxin-associated illness: multiple-system symptoms, a vision deficit, and effect of cholestyramine. Neurotoxicol Teratol. 2005;27(5):733-743.

7. Shoemaker RC, House DE. A time-series study of sick building syndrome: chronic, biotoxin-associated illness from exposure to water-damaged buildings. Neurotoxicol Teratol. 2005;27(1):29-46.

8. Shoemaker RC, House DE. Sick building syndrome (SBS) and exposure to water-damaged buildings: time series study, clinical trial and mechanisms. Neurotoxicol Teratol. 2006;28(5):573-588.

9. McMahon SW, Kundomal KA, Yangalasetty S. Pediatrics norms for visual contrast sensitivity using an APT VCS tester. Med Res Arch. 2017;5(5).

10. Pelli DG, Bex P. Measuring contrast sensitivity. Vision Res. 2013;90:10-14.

11. Owsley C. Contrast sensitivity. Ophthalmol Clin North Am. 2003;16(2):171-177.

12. Thomas G, Burton NC, Mueller C, Page E, Vesper S. Comparison of work-related symptoms and visual contrast sensitivity between employees at a severely water-damaged school and a school without significant water damage. Am J Ind Med. 2012;55(9):844-854. doi:10.1002/ajim.22059

13. Jimenez-Barbosa IA, Di Lizio S, Ahn SB, et al. Assessment of visual contrast sensitivity in biotoxin-exposed individuals using four testing methods. Ann Med. 2026;58(1):2646821. doi:10.1080/07853890.2026.2646821

14. American College of Medical Toxicology. Position statement: medical toxicology considerations in the diagnosis and treatment of patients with concerns about mold-related inhalation exposures. 2025.

15. Denning DW, Chakrabarti A. Pulmonary and sinus fungal diseases in non-immunocompromised patients. Lancet Infect Dis. 2017;17(11):e357-e366. doi:10.1016/S1473-3099(17)30309-2

16. Valtonen V. Clinical diagnosis of the dampness and mold hypersensitivity syndrome: review of the literature and suggested diagnostic criteria. Front Immunol. 2017;8:951. doi:10.3389/fimmu.2017.00951